

Panel Data Methods

Lecture 3: Pooled OLS, First Differences, Fixed Effects,
Random Effects

Where We Are

Recap from last lecture

Difference-in-differences:

- Compare *changes* across treated and control groups
- Identifying assumption: parallel trends
- Workhorse for policy evaluation when randomization isn't possible

DiD requires only *two* time periods (pre and post) and *two* groups (treated and control). Today: what if we have more units, more time periods, and richer variation?

Panel data.

Today's roadmap

1. What is panel data? How does it differ from cross-sections and time-series?
2. Why we usually *need* panel methods: omitted variable bias
3. The unobserved effects model
4. Why pooled OLS fails
5. First differencing
6. Fixed effects (within estimator)
7. Random effects
8. Random vs. Fixed: the Hausman test and intuitive reasoning
9. Two-way fixed effects: connecting back to DiD

Three Kinds of Data

Cross-sectional data

A **cross-section** is a snapshot: many units observed at one point in time.

Example: The 2024 Canadian Labour Force Survey records wages, education, age, and occupation for thousands of workers. Each row is a different worker; the year is fixed.

Notation:

$$y_i = \beta_0 + \beta_1 x_i + u_i, \quad i = 1, \dots, N$$

We compare *across units* at a single moment.

Time series data

A **time series** is one unit observed repeatedly over time.

Example: Canadian GDP, quarterly, 1961–2024. One country, many quarters.

Notation:

$$y_t = \beta_0 + \beta_1 x_t + u_t, \quad t = 1, \dots, T$$

We compare *within a single unit, across time*.

Panel data (longitudinal data)

Panel data combines both: the same units, observed at multiple time periods.

Example: The Panel Study of Income Dynamics (PSID) has followed ~5,000 American families since 1968, asking about income, employment, health each year.

Notation — now *two* indices:

$$y_{it} = \beta_0 + \beta_1 x_{it} + u_{it}, \quad i = 1, \dots, N; \ t = 1, \dots, T$$

A balanced panel has $N \cdot T$ observations.

Panel data vs. pooled cross-sections

Crucial distinction. Pooled cross-sections stack different surveys; panel data follows the same units.

	Cross section	Time series	Pooled cross-sec.	Panel data
Multiple units	✓		✓	✓
Multiple periods		✓	✓	✓
Same units over time		(just one)		✓

Only panel data lets us control for **unobserved, time-invariant characteristics** of a unit. This is the single most important feature of panel data.

Why this matters: a marriage premium

Question: does marriage increase men's wages?

With a cross-section:

Compare married men to unmarried men in 2024. But married men may differ in unobserved ways — more disciplined, more ambitious, more responsible.

With pooled cross-sections (e.g., 2014 and 2024):

Same problem, twice.

With panel data:

Watch *the same man* before and after he marries. Whatever stable trait makes him him — discipline, intelligence, family background — is held constant.

That is the power of the panel.

Practical note: N vs. T in economic data

In most economic applications:

- N (number of units) is large: thousands of individuals, hundreds of firms, dozens of countries
- T (number of periods) is small: 2–20 years is typical

This is called a **short panel** or *large- N , small- T* setting.

Why does this matter?

- Asymptotic theory typically takes $N \rightarrow \infty$ with T fixed
- Time-series methods (autoregression, unit roots) often less relevant
- Standard errors and inference are designed around this regime

(*Long panels* — macro time-series for many countries, financial returns over decades — need different tools.)

The Running Example: Housing Prices and Crime

Motivating question

Do crime rates affect housing prices?

Setting:

- We observe many US cities ($i = 1, \dots, N$)
- Over several years ($t = 1, \dots, T$)
- For each city-year, we record:
 - HP_{it} : average housing price
 - $crime_{it}$: crime rate (offenses per 1,000 residents)

Hypothesis: higher crime $\Rightarrow$ lower demand for housing $\Rightarrow$ lower prices.

But this is harder than it looks.

What forces are at play?

Housing prices in city i at year t depend on many things:

City-specific, time-invariant factors:

- Geography (coastal, mountainous)
- Long-run demographic composition
- Average education level
- Climate
- Distance to economic centers

Year-specific, common to all cities:

- National income trends
- Federal interest rates
- Business cycle conditions
- Mortgage market regulations

Idiosyncratic shocks unique to a specific city-year.

The full model

We can write a regression that explicitly accounts for these three sources of variation:

$$HP_{it} = \beta_0 + \beta_1 \cdot crime_{it} + v_t + a_i + e_{it}$$

Where:

- β_1 : **the causal parameter** we want — the effect of crime on prices
- v_t : **time-specific factors** — common to all cities in year t (rising national income, federal policy)
- a_i : **city-specific factors** — constant over time within city i (education levels, geography, demographics)
- e_{it} : **idiosyncratic error** — city-year specific shocks, mean-zero

Decomposing the error: an important shift

A regular regression has *one* error term:

$$y_i = \beta_0 + \beta_1 x_i + u_i$$

Panel data invites us to *split* the error into pieces with different structure:

$$\text{error}_{it} = \underbrace{v_t}_{\text{time-specific}} + \underbrace{a_i}_{\text{city-specific}} + \underbrace{e_{it}}_{\text{idiosyncratic}}$$

Why split it?

- Each piece has different properties (constant vs. varying)
- Each piece has a different relationship with our regressor
- Each piece needs a different solution

This decomposition is the key conceptual move of the entire lecture.

What's in a_i ? (city-specific, time-invariant)

Things that don't change much over a few years for city i :

- **Education levels.** Average years of schooling shifts slowly. Cities with high education levels tend to have less crime *and* higher housing prices.
- **Geography.** A coastal city is always coastal. Distance to a major port doesn't change.
- **Long-run demographic composition.** Age distribution, ethnic composition — these drift, but slowly.
- **Historical institutions.** The legal regime, local policing culture, school quality.

These are *exactly* the variables that confound the crime→prices relationship in a single cross-section.

What's in v_t ? (time-specific, common to all cities)

Things that affect every city in year t :

- **Macro conditions.** National GDP growth, recessions.
- **Federal monetary policy.** Interest rates affect mortgages everywhere.
- **Federal fiscal policy.** Tax changes affecting after-tax income.
- **National-level regulatory changes.** Banking regulations, environmental rules.

These cause every city's housing prices to drift together — but they're not what we want to attribute to crime.

What's in e_{it} ? (idiosyncratic)

City-year-specific shocks — the genuinely random component:

- A major employer opens or closes a plant
- A natural disaster (flood, fire)
- A high-profile local event affecting reputation
- Measurement error in housing prices or crime statistics

We assume $E[e_{it}] = 0$ and that e_{it} is uncorrelated with $crime_{it}$.

This is the only “error” we’d have in an ideal world. The trouble comes from a_i and v_t .

Time Dummies and the Dummy Variable Trap

How do we handle v_t ?

The time-specific effects v_t shift the intercept of our regression up or down each year.

We can absorb them by introducing **time dummy variables** — one for each year except a base year:

$$HP_{it} = \beta_0 + \beta_1 crime_{it} + \gamma_1 \delta_{2t} + \gamma_2 \delta_{3t} + \cdots + \gamma_{T-1} \delta_{Tt} + a_i + e_{it}$$

Where:

$$\delta_{kt} = \begin{cases} 1 & \text{if } t = k \\ 0 & \text{otherwise} \end{cases}$$

We've replaced v_t (a continuous unobserved variable) with $T - 1$ explicit dummies.

The dummy variable trap

Why $T - 1$ dummies and not T ?

If we included one dummy for every year:

$$\delta_{1t} + \delta_{2t} + \delta_{3t} + \cdots + \delta_{Tt} = 1 \text{ for every observation}$$

That's perfectly collinear with the intercept β_0 (which is also $= 1$ for every observation). OLS cannot identify both.

Solution: drop one dummy (say, δ_{1t} , for year 1) to serve as the **base year**.

This isn't an arbitrary loss — the base year is absorbed into the intercept β_0 .

Interpreting the time dummies

With year 1 as the base year:

- γ_1 : average difference in housing prices in year 2 vs. year 1, holding crime constant
- γ_2 : average difference in year 3 vs. year 1
- ...
- γ_{T-1} : average difference in year T vs. year 1

The γ coefficients trace out the common time path of housing prices, after accounting for crime.

Concrete example. If $\gamma_3 = 5000$ (and prices are in dollars), then on average across all cities, year 4's housing prices are \$5,000 higher than year 1's — after accounting for crime rate differences.

Why not similar treatment for cities?

You might think: “okay, drop a city dummy, include $N - 1$ city dummies, problem solved.”

This is exactly the LSDV (least squares dummy variable) approach to fixed effects — we’ll get there in section 6.

But first, what happens if we *ignore* the city-specific effects a_i entirely and just run OLS? Let’s see why that fails.

Why Pooled OLS Fails

Pooled OLS: treat panel data as a giant cross-section

The simplest thing to do with panel data is to ignore the panel structure — just stack all $N \cdot T$ observations and run OLS:

$$HP_{it} = \beta_0 + \beta_1 crime_{it} + u_{it}$$

Where:

$$u_{it} = \alpha_i + e_{it}$$

(For simplicity, let's set aside v_t for now — assume we've already absorbed time effects with dummies.)

Each city-year is treated as an independent observation. This is **pooled OLS**.

The consistency requirement

Recall the OLS consistency condition: the regressor must be uncorrelated with the error.

$$\text{Cov}(u_{it}, \text{crime}_{it}) = 0$$

Plugging in our decomposition:

$$\text{Cov}(u_{it}, \text{crime}_{it}) = \text{Cov}(a_i + e_{it}, \text{crime}_{it})$$

Expanding:

$$\text{Cov}(a_i + e_{it}, \text{crime}_{it}) = \text{Cov}(a_i, \text{crime}_{it}) + \text{Cov}(e_{it}, \text{crime}_{it})$$

Where the bias comes from

$$\text{Cov}(a_i + e_{it}, \text{crime}_{it}) = \text{Cov}(a_i, \text{crime}_{it}) + \text{Cov}(e_{it}, \text{crime}_{it})$$

Even if we're willing to assume the idiosyncratic shock is exogenous:

$$\text{Cov}(e_{it}, \text{crime}_{it}) = 0$$

We're still left with:

$$\text{Cov}(a_i, \text{crime}_{it}) = ?$$

This is the part that should worry us.

Why $\text{Cov}(a_i, \text{crime}_{it}) \neq 0$

Recall what's in a_i : education levels, geography, demographic composition. . .

Are these likely to be uncorrelated with crime?

Education:

- Cities with higher average education tend to have less crime
- $\Rightarrow \text{Cov}(\text{education in } a_i, \text{crime}_{it}) < 0$

Geography:

- Cities with certain geographic features (urban density, port access) have systematically different crime profiles

$\Rightarrow \text{Cov}(a_i, \text{crime}_{it}) \neq 0$, and the pooled OLS estimator $\hat{\beta}_1^{POLS}$ is **inconsistent**.

Even when consistent, the SEs are wrong

A second problem with pooled OLS: the errors $u_{it} = a_i + e_{it}$ for the same city in different years are **correlated**:

$$\text{Cov}(u_{it}, u_{is}) = \text{Cov}(a_i + e_{it}, a_i + e_{is}) = \text{Var}(a_i) \neq 0 \quad \text{for } t \neq s$$

Each city's observations across years share the city-specific component a_i .

Standard OLS standard errors assume independent observations $\Rightarrow$ they will be **too small**, leading to overconfident inference.

Practical fix: cluster standard errors at the unit (city) level.

But this doesn't solve the bias — only the inference. We still need a different estimator.

Summary: pooled OLS

- **Easy:** just run OLS on the stacked data
- **Biased:** whenever a_i is correlated with regressors (almost always in practice)
- **Misleading inference:** naive standard errors are too small

When useful?

- As a baseline comparison
- When you genuinely believe a_i is uncorrelated with x — but then random effects is probably better (we'll see why)

We need methods that explicitly handle a_i .

First Differencing

The first-differencing idea

a_i is constant over time. So if we take differences between consecutive periods, a_i should drop out.

Original model at time t :

$$HP_{it} = \beta_0 + \beta_1 crime_{it} + a_i + e_{it}$$

At time $t - 1$:

$$HP_{i,t-1} = \beta_0 + \beta_1 crime_{i,t-1} + a_i + e_{i,t-1}$$

Subtract:

$$\underbrace{HP_{it} - HP_{i,t-1}}_{\Delta HP_{it}} = \beta_1 \underbrace{(crime_{it} - crime_{i,t-1})}_{\Delta crime_{it}} + \underbrace{(e_{it} - e_{i,t-1})}_{\Delta e_{it}}$$

$$\Delta HP_{it} = \beta_1 \Delta crime_{it} + \Delta e_{it}$$

Key observation: a_i appears in both periods, with the same value.
When we subtract, it cancels:

$$a_i - a_i = 0$$

The intercept also cancels: $\beta_0 - \beta_0 = 0$.

We can now run OLS on ΔHP_{it} regressed on $\Delta crime_{it}$, and the city-specific bias is eliminated.

This is the **first-difference (FD)** estimator.

Interpretation: change-on-change

$\hat{\beta}_1^{FD}$ answers:

“When a city’s crime rate changes from one year to the next, how does that city’s housing price change?”

This is a **change-on-change** interpretation.

Concrete example.

- Boston’s crime rate fell by 5 offenses per 1,000 between 2010 and 2011
- Boston’s housing prices rose by \$8,000 over the same period
- This is one data point for the FD regression

We’re using **within-city variation over time** to identify β_1 .

What FD requires: strict exogeneity

For $\hat{\beta}_1^{FD}$ to be consistent:

$$\text{Cov}(\Delta \text{crime}_{it}, \Delta e_{it}) = 0$$

This requires the idiosyncratic shock to be unrelated to changes in crime — not just in period t , but across periods.

Formally, we need:

$$E[e_{it} | \text{crime}_{i1}, \text{crime}_{i2}, \dots, \text{crime}_{iT}, a_i] = 0 \quad \forall t$$

This is called **strict exogeneity**. We'll discuss it more later.

Notice: **no assumption** about the relationship between a_i and crime_{it} . We've differenced a_i away.

What FD assumes about effects: immediacy

What are we assuming about the effects of education using first-difference identification? **That the effects of the “shock” are immediate.**

Wage–education thought experiment. Suppose wage and education have this one-to-one relationship:

wage:	5	5	5	6	6	6	7	7	7
educ:	7	7	7	8	8	8	9	9	9

In first differences:

Δ wage:	0	0	0	1	0	0	1	0	0
Δ educ:	0	0	0	1	0	0	1	0	0

$\Rightarrow$ correlation = 1. FD recovers the relationship perfectly.

What if effects are lagged?

Now suppose the wage responds with a one-year lag:

wage:	0	0	0	0	1	0	0	1	0
educ:	0	0	0	1	0	0	1	0	0

$\Rightarrow$ correlation = $-0.286!$

First differencing evaluates short-run effects. If the underlying causal effect takes time to materialize, FD can give misleading estimates — even reversing the sign.

The problem: Δx_{it} is contemporaneous, but the effect on y doesn't arrive until $t + 1$.

FD handles gradual effects better

What if changes are gradual rather than one-shot?

wage:	5	5	5	6	7	8	8	8	8
educ:	6	6	6	7	8	9	9	9	9

In first differences:

Δ wage:	0	0	0	1	1	1	0	0	0
Δ educ:	0	0	0	1	1	1	0	0	0

$\Rightarrow$ correlation = 1 if effects are immediate.

With one-year lag $\Rightarrow$ correlation = 0.5 (still informative).

Lagged effects are less of an issue when the underlying effects are gradual.

Practical considerations for FD

- **Lose one period per unit.** With T periods, we get $T - 1$ differences. With $T = 2$, we get a cross-section of changes.
- **Time-invariant variables drop out.** Race, sex, country of birth all disappear in differences (just as a_i did).
- **Sensitive to lagged effects.** As we just saw.

Two solutions to lagged-effects sensitivity:

1. Take *long differences* (e.g., $y_{i,T} - y_{i,1}$)
2. Use a *within estimator* — i.e., fixed effects

This brings us to the workhorse method.

Fixed Effects (Within Estimator)

The fixed effects idea

First differencing eliminated a_i by subtracting consecutive periods.

Fixed effects eliminates a_i by subtracting each unit's *time average*.

Start with:

$$HP_{it} = \beta_0 + \beta_1 crime_{it} + a_i + e_{it} \quad (1)$$

For each city i , average over time:

$$\bar{HP}_i = \beta_0 + \beta_1 \overline{crime}_i + a_i + \bar{e}_i \quad (2)$$

Note: a_i appears unchanged in (2) because it doesn't vary with t .

The within transformation

Subtract (2) from (1):

$$HP_{it} - \bar{HP}_i = \beta_1(\text{crime}_{it} - \overline{\text{crime}_i}) + (e_{it} - \bar{e}_i)$$

Where the a_i and β_0 terms cancel:

$$a_i - a_i = 0, \quad \beta_0 - \beta_0 = 0$$

Define **time-demeaned variables**:

$$\ddot{HP}_{it} = HP_{it} - \bar{HP}_i, \quad \ddot{\text{crime}}_{it} = \text{crime}_{it} - \overline{\text{crime}_i}, \quad \ddot{e}_{it} = e_{it} - \bar{e}_i$$

The transformed model:

$$\ddot{HP}_{it} = \beta_1 \ddot{\text{crime}}_{it} + \ddot{e}_{it}$$

The within estimator

Running OLS on:

$$\ddot{HP}_{it} = \beta_1 \ddot{crime}_{it} + \ddot{e}_{it}$$

gives the **fixed effects estimator** (a.k.a. the **within estimator**):

$$\hat{\beta}_1^{FE} = \frac{\sum_i \sum_t \ddot{crime}_{it} \cdot \ddot{HP}_{it}}{\sum_i \sum_t \ddot{crime}_{it}^2}$$

Key insight: we're using only the variation *within each city, around its own mean*.

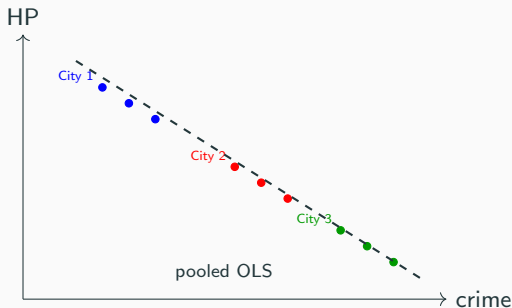
Between-city variation has been thrown away.

What the within transformation does, intuitively

Imagine plotting all $N \cdot T$ data points as $(crime_{it}, HP_{it})$.

Pooled OLS fits a single line through the cloud.

The within transformation shifts every city's data so that each city's mean sits at the origin. Then it fits a line through the recentered cloud.



Pooled OLS reflects both within and between variation. FE keeps only the

Numerical example: FE vs. FD with lagged effects

Return to the wage–education sequence with one-year lagged response:

wage:	5	5	5	5	6	6	6	7	7
educ:	7	7	7	8	8	8	9	9	9

First differences:

correlation = -0.286 (badly misleading)

Within transformation (subtract sample mean of each):

<i>w</i> äge:	-0.78	-0.78	-0.78	-0.78	0.22	0.22	0.22	1.22	1.22
<i>e</i> duc:	-1.00	-1.00	-1.00	0.00	0.00	0.00	1.00	1.00	1.00

correlation = 0.866 — much more informative.

FE captures average effects within each unit; FD only captures contemporaneous changes.

Equivalent approach: LSDV (dummies for each unit)

There's a second way to think about fixed effects: just add a dummy for every unit (except one):

$$HP_{it} = \beta_0 + \beta_1 \text{crime}_{it} + \alpha_2 D_{2i} + \alpha_3 D_{3i} + \cdots + \alpha_N D_{Ni} + e_{it}$$

Where $D_{ji} = 1$ if observation belongs to city j , 0 otherwise.

This is called the **Least Squares Dummy Variable (LSDV) estimator**.

Key fact: LSDV and the within estimator produce *numerically identical* slope coefficients.

(Frisch–Waugh–Lovell theorem.)

Why the equivalence?

Both methods accomplish the same thing: they let each unit have its own intercept.

- Demeaning: subtract each unit's average $\Rightarrow$ identifies slopes from within-unit variation
- LSDV: include a dummy per unit $\Rightarrow$ each dummy absorbs that unit's level

Which to use?

- **Within** (demeaning): computationally cheap when N is huge
- **LSDV** (dummies): conceptually clearer for students; infeasible when N is very large

Modern software uses the within transformation under the hood — but the dummy-variable picture often makes “fixed effects” click.

What variation identifies the FE coefficients?

Only **within-unit variation over time**.

Two important consequences:

1. Time-invariant variables cannot be estimated.

- A city's coastal status (1 if coastal, 0 otherwise) doesn't vary over time within the city
- Demeaning: $\text{coastal}_i - \text{coastal}_i = 0$ for every observation
- There's no variation left to estimate a coefficient

2. The estimator effectively answers:

"When city i 's crime rate changes, how much does city i 's housing price change?"

The killer trade-off

This is the central trade-off of fixed effects:

What we gain:

- Eliminates bias from *all* time-invariant unobserved confounders — whether we can name them or not
- Education, geography, demographic composition, historical institutions: all absorbed by a_i

What we lose:

- Cannot estimate effects of any variable that doesn't change within unit
- Lose statistical power compared to using both between and within variation

For most causal questions in microeconomics, the trade-off is worth it.

The key assumption: strict exogeneity

For FE to be consistent, we need:

$$E[e_{it} | crime_{i1}, crime_{i2}, \dots, crime_{iT}, a_i] = 0 \quad \forall t$$

This is **strict exogeneity** (conditional on a_i).

What this says intuitively:

Given the city's stable characteristics a_i , the idiosyncratic shock at time t is unrelated to crime rates in *any* period — past, present, or future.

What this is NOT:

Strict exogeneity is *stronger* than $E[e_{it} | crime_{it}, a_i] = 0$. It rules out feedback between past shocks and future regressors.

What strict exogeneity rules out: feedback

Example violation. Suppose this year's housing price shock e_{it} affects next year's crime:

- A negative price shock causes wealthier residents to leave city
- Population composition shifts toward higher crime $\Rightarrow crime_{i,t+1}$ rises

Then e_{it} is correlated with $crime_{i,t+1}$ — violating strict exogeneity.

Why this matters: FE assumes regressors are “exogenous in every period.” When current shocks feed into future regressors, this fails.

This is a real worry in many economic settings. Dynamic panel methods (Arellano–Bond) handle this, but they're beyond third-year material.

Notice: strict exogeneity says *nothing* about a_i . We've already taken care of a_i by differencing/demeaning.

When is FE preferred over FD?

Both eliminate a_i . Both require strict exogeneity. So when does each do better?

If e_{it} is...	More efficient:
Uncorrelated over time (serially independent)	Fixed effects
A random walk (highly persistent)	First differences

Practical advice: when in doubt, run both. If they agree, you're in good shape. If they disagree substantially, investigate.

Special case: when $T = 2$, FE and FD are numerically identical.

Degrees of freedom

With N units and T periods, the panel has NT observations.

But the within transformation costs us N degrees of freedom — one for each unit-specific mean we subtracted.

Plus k for the slope coefficients estimated.

So:

$$df = NT - N - k$$

This matters for standard errors and t -statistics. Software handles it automatically — but it's worth knowing where the adjustment comes from.

Random Effects

The motivation for random effects

We've spent a lot of effort eliminating a_i .

But what if a_i is *not* correlated with our regressors?

In that case:

- There is no bias from a_i to begin with
- Eliminating a_i via FE throws away useful information — the between-unit variation
- We could do *better* (more efficient) by keeping that variation

Random effects is the model that exploits this.

The random effects assumption

Random effects (RE) assumption:

$$E[a_i | crime_{i1}, crime_{i2}, \dots, crime_{iT}] = 0$$

In plain English: the unobserved effect a_i is just random noise, drawn for each unit independently of the regressors. It's like another error term.

Compare to FE:

- RE: a_i is uncorrelated with x_{it} (strong assumption)
- FE: makes *no assumption* about the relationship between a_i and x_{it}

RE is more demanding. If the assumption holds, we gain efficiency. If it fails, we get bias.

When is the RE assumption plausible?

Plausible:

- Randomly sampled units where the regressor is randomly assigned (e.g., a clinical trial with random dose assignment)
- Settings where the unobserved heterogeneity really is “luck of the draw” — unrelated to treatment

Implausible:

- Our housing/crime example: cities with high education (a_i) have systematically lower crime
- Wage/education: ability (a_i) correlates with schooling
- Firm productivity: managerial talent (a_i) correlates with capital and labor choices

In most observational economic data, RE is a stretch.

The RE estimator: quasi-demeaning

The RE estimator runs OLS on a *partially* demeaned equation:

$$HP_{it} - \lambda \bar{HP}_i = \beta_0(1 - \lambda) + \beta_1(\text{crime}_{it} - \lambda \overline{\text{crime}_i}) + (v_{it} - \lambda \bar{v}_i)$$

Where λ is a weight between 0 and 1:

$$\lambda = 1 - \sqrt{\frac{\sigma_e^2}{\sigma_e^2 + T\sigma_a^2}}$$

$$\sigma_a^2 = \text{Var}(a_i) \text{ and } \sigma_e^2 = \text{Var}(e_{it}).$$

We *partially* subtract each unit's mean, rather than fully (as in FE) or not at all (as in pooled OLS).

The intuition of λ

λ tells us how much of the between-unit variation to throw away.

Two limits:

Case 1: $\sigma_a^2 = 0$ (no unobserved heterogeneity)

$$\lambda = 1 - \sqrt{\frac{\sigma_e^2}{\sigma_e^2}} = 0$$

$\Rightarrow$ no demeaning $\Rightarrow$ RE collapses to **pooled OLS**.

Case 2: $\sigma_a^2 \rightarrow \infty$ (huge unobserved heterogeneity)

$$\lambda \rightarrow 1 - 0 = 1$$

$\Rightarrow$ full demeaning $\Rightarrow$ RE collapses to **fixed effects**.

RE is a compromise between pooled OLS and FE.

The advantages of RE (when its assumption holds)

1. Time-invariant variables can be estimated.

- Because we don't fully demean, variables like coastal status, race, or sex aren't wiped out
- Their coefficients can be estimated

2. More efficient than FE.

- Uses both within-unit and between-unit variation
- Lower standard errors when the RE assumption is correct

3. Properly handles the serial correlation in $u_{it} = a_i + e_{it}$.

- Standard RE software uses GLS to account for the correlation structure

The catch

If a_i is correlated with regressors, RE is biased and inconsistent.

And this correlation is *exactly* the situation we use panel data to handle!

So in many applied economic settings:

- Theory tells us a_i is likely correlated with regressors
- RE's identifying assumption fails
- RE is biased
- FE is the safer choice

But how do we *know* whether the RE assumption holds in our data?

The Hausman test.

Random vs. Fixed Effects

The trade-off, visualized

	FE	RE
Assumption on a_i	None (anything goes)	Uncorrelated with x
Consistency	Always	Only if RE assumption holds
Efficiency	Lower	Higher
Time-invariant x	Can't estimate	Can estimate

Summary:

- RE: gambles on a strong assumption for a gain in efficiency
- FE: pays a small efficiency cost for unconditional robustness

We want to choose between these based on whether the gamble is worth it.

Intuitive example: when RE is plausible

Setting: a medical trial randomly assigns drug doses to patients. Each patient is followed over time.

- a_i = patient-specific traits (genetics, baseline health)
- x_{it} = drug dose at visit t

Is $\text{Cov}(a_i, x_{it}) = 0$?

Yes, by design. Doses are randomly assigned — they're statistically independent of patient genetics.

Implication: RE is consistent here. It's also more efficient than FE. Use RE.

Intuitive example: when RE fails

Setting: our housing/crime example.

- a_i = city's average education level, geography, demographic composition
- $crime_{it}$ = crime rate

Is $Cov(a_i, crime_{it}) = 0$?

Very unlikely. Cities with higher education *systematically* have lower crime. The components of a_i are exactly what determines $crime_{it}$.

Implication: RE is biased here. We should use FE.

General principle: when units are not randomly assigned to “treatment” (here, levels of crime), a_i usually correlates with regressors $\Rightarrow$ FE is safer.

The Hausman test: the logic

We have two estimators:

- **FE**: consistent always
- **RE**: consistent only if a_i is uncorrelated with regressors

Under the RE assumption:

- Both are consistent
- They estimate the same β_1
- Their estimates should be similar in large samples

Under correlation between a_i and x :

- FE is still consistent
- RE is not — it's biased toward something else
- Their estimates will diverge

So: compare $\hat{\beta}^{FE}$ and $\hat{\beta}^{RE}$. If they're far apart, that's evidence that RE is misspecified.

The Hausman test statistic

$$H = \left(\hat{\beta}_{FE} - \hat{\beta}_{RE} \right)' \left[\hat{V}_{FE} - \hat{V}_{RE} \right]^{-1} \left(\hat{\beta}_{FE} - \hat{\beta}_{RE} \right)$$

Where:

- $\hat{\beta}_{FE} - \hat{\beta}_{RE}$ is the vector of coefficient differences
- $\hat{V}_{FE} - \hat{V}_{RE}$ is the difference of the variance-covariance matrices

Under the null hypothesis (RE is correct), $H \sim \chi_k^2$ where k is the number of time-varying regressors.

Decision rule:

- Reject the null (large H , small p) $\Rightarrow$ RE is inconsistent. Use FE.
- Fail to reject $\Rightarrow$ RE may be fine. Use RE for efficiency.

Cautions about Hausman

1. Underpowered in small samples.

- Failing to reject doesn't *prove* RE is correct
- Just means we couldn't detect the bias

2. Assumes FE is consistent.

- If strict exogeneity fails, FE is also biased, and the test is meaningless

3. The test is a guide, not a verdict.

- Theory matters: if you have good reason to think a_i is correlated with x , prefer FE *regardless* of what Hausman says
- In most applied micro, FE is the default and Hausman is a sanity check

The applied micro consensus

Step 1: Theory check.

- Is there any reason to think a_i might correlate with my regressors?
- For most observational economic data: yes.
- $\Rightarrow$ Lean toward FE.

Step 2: Statistical check (Hausman).

- Rejection: confirms FE is needed.
- Non-rejection: doesn't necessarily mean RE is fine — the test is underpowered.

Step 3: Robustness.

- Show both estimates. If they agree, the choice doesn't matter much.
- If they disagree, FE is the safer report.

In practice: FE is the default unless you have a strong reason otherwise.

Two-Way Fixed Effects and DiD

Adding time effects back in

We've eliminated a_i . But what about v_t ?

The natural extension: include *both* unit fixed effects and time fixed effects:

$$HP_{it} = \beta_0 + \beta_1 crime_{it} + a_i + d_t + e_{it}$$

Where:

- a_i : city fixed effect — absorbs all time-invariant city characteristics
- d_t : year fixed effect — absorbs all shocks common to every city in year t

This is called **two-way fixed effects (TWFE)**.

What TWFE identifies

We use only variation *within a city, within a time period*.

Specifically, TWFE absorbs:

- Any stable trait of a city (education, geography, demographic composition)
- Any shock that hit every city in a given year (recessions, interest rate changes, federal policy)

What's left: variation in crime within city i , after netting out the city's average path and the national year-by-year trend.

TWFE is the modern default in applied microeconometrics.

Implementing TWFE

Approach 1: Two-way demeaning.

$$\ddot{HP}_{it} = HP_{it} - \bar{HP}_i - \bar{HP}_t + \bar{HP}$$

(Subtract city mean, subtract year mean, add back grand mean.)

Approach 2: Dummies for everyone.

$$HP_{it} = \beta_1 crime_{it} + \sum_{j=2}^N \alpha_j D_{ji} + \sum_{k=2}^T \gamma_k \delta_{kt} + e_{it}$$

Approach 3: Modern econometric software.

In Stata: `xtreg y x i.year, fe`

In R: `feols(y ~ x | unit + year, data = df)` (from `fixest`)

All three produce identical slope estimates.

Connecting back to DiD

Difference-in-differences is a special case of two-way fixed effects.

Recall the DiD specification from last lecture:

$$wage_{it} = \beta_0 + \beta_1 Post_t + \beta_2 Treat_i + \delta(Post_t \cdot Treat_i) + u_{it}$$

Here:

- $Treat_i$ acts like a (simple) unit fixed effect: 1 if treated, 0 if not
- $Post_t$ acts like a (simple) time fixed effect: 1 if post-period, 0 if pre
- δ is the DiD coefficient

With more groups and more periods, $Treat_i$ becomes a full set of unit dummies, $Post_t$ becomes a full set of year dummies, and the interaction becomes the TWFE-equivalent of DiD.

Reading the DiD regression as TWFE

	Before	After	After - Before
Control	β_0	$\beta_0 + \beta_1$	β_1
Treatment	$\beta_0 + \beta_2$	$\beta_0 + \beta_1 + \beta_2 + \delta$	$\beta_1 + \delta$
Control - Treatment	β_2	$\beta_2 + \delta$	δ

The 2×2 table is exactly what TWFE does with $N = 2$ groups and $T = 2$ periods:

- Control before: β_0 (the baseline)
- Treatment before: $\beta_0 + \beta_2$ (β_2 is the “treatment FE”)
- Control after: $\beta_0 + \beta_1$ (β_1 is the “year FE”)
- Treatment after: $\beta_0 + \beta_1 + \beta_2 + \delta$ (δ is the DiD)

So what does TWFE buy us beyond DiD?

TWFE is the generalization of DiD to:

- Many treated units, not just one
- Many time periods, not just pre/post
- Staggered treatment timing (different units treated at different times)
- Continuous treatments (not just binary)

Modern empirical work (Card minimum wage papers, Clemens & Strain, Addison et al.) all use TWFE-type specifications.

(Recent literature has identified subtle issues with TWFE under staggered treatment — Goodman-Bacon 2021, de Chaisemartin & D'Haultfœuille 2020 — which we'll touch on later. For now: TWFE is the default.)

Putting It All Together

The decision tree

When you face panel data, ask:

1. Is a_i plausibly correlated with my regressors?

- Yes (almost always in applied econ) $\Rightarrow$ FE or FD
- No $\Rightarrow$ consider RE for efficiency; test with Hausman

2. Do I care about time-invariant variables?

- Yes $\Rightarrow$ FE/FD won't estimate them. Use RE or hybrid models.
- No $\Rightarrow$ FE is fine.

3. Is strict exogeneity reasonable?

- Yes $\Rightarrow$ FE/FD/RE all valid (subject to point 1).
- No (feedback) $\Rightarrow$ need dynamic panel methods (beyond this course).

4. How many time periods do I have?

- $T = 2$: FE and FD are identical; pick whichever's more intuitive
- $T > 2$: think about error structure (serial correlation $\Rightarrow$ FE; random walk $\Rightarrow$ FD)

5. Should I include time effects?

- Almost always yes $\Rightarrow$ TWFE

6. Inference?

- Always cluster standard errors at the unit level
- Even more conservative: cluster at the unit and time level

Summary table

Method	Removes a_i ?	Time-invariant x ?	Key assumption	When
Pooled OLS	No	Yes	$Cov(x, a_i) = 0$	Baseline on
Random effects	Partially	Yes	$Cov(x, a_i) = 0$	Random ment
Fixed effects	Yes	No	Strict exogeneity	Default
First differences	Yes	No	Strict exogeneity	FE alt., T same
LSDV	Yes (explicit)	No	Same as FE	Equiv. to FE
Two-way FE	Yes + time	No	Strict exogeneity	Modern def

Core message: panel data lets us control for unobserved time-invariant heterogeneity. That's a remarkable capability. The cost is that we identify from within-unit variation only.

Common pitfalls

1. **Forgetting that time-invariant variables drop out of FE.** Wanting the effect of sex on wages with FE returns nothing — sex doesn't vary within person.
2. **Using non-clustered standard errors.** The composite error is serially correlated. Always cluster at the unit level.
3. **Confusing FD and FE.** Not the same except when $T = 2$.
4. **Interpreting FE coefficients as cross-sectional comparisons.** They're not. They are within-unit changes.
5. **Treating Hausman as a black box.** Combine the test with economic reasoning.
6. **Ignoring strict exogeneity violations.** Feedback from past shocks to current regressors invalidates FE/FD.

Recap

1. Panel data = same units, multiple time periods.
2. The unobserved effect a_i is the central feature: it captures time-invariant heterogeneity.
3. Pooled OLS is biased when a_i correlates with regressors. Almost always does.
4. **First differences** and **fixed effects** both eliminate a_i . FE is more efficient with serially independent errors; FD with persistent ones.
5. **Random effects** is more efficient when a_i is uncorrelated with regressors — but biased when it isn't.
6. **Hausman test** formalizes the FE vs. RE choice. Use it alongside theory.
7. **TWFE** is the modern default — and DiD is a special case.

Regression discontinuity design.

Until now, we've handled selection bias by:

- Randomizing (RCTs, A/B tests)
- Comparing changes (DiD)
- Within-unit variation (panel methods)

Next: a fourth approach. Use a sharp cutoff rule — where being just above or just below the threshold is essentially random — to identify causal effects.

Reading: Wooldridge Ch. 13–14 (panel data) and Stock & Watson Ch. 10.

References

- Wooldridge, J. M. (2013). *Introductory Econometrics: A Modern Approach*. Chs. 13–14.
- Stock, J. H. & Watson, M. W. (2009). *Introduction to Econometrics*. Ch. 10.
- Angrist, J. D. & Pischke, J.-S. (2009). *Mostly Harmless Econometrics*. Ch. 5.
- Hausman, J. A. (1978). Specification tests in econometrics. *Econometrica*, 46(6), 1251–1271.
- Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of Econometrics*, 225(2), 254–277.